

Module 4

Two-Port Networks and Network Graphs

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Fourier Transform

The Fourier transform converts a time-domain function into a **frequency-domain** function while the Laplace transform converts it into an **s-domain** function.

This means the Laplace transform can analyze a broader range of functions, including those that are unstable or grow over time, due to its inclusion of the real part () which represents exponential decay or growth, unlike the Fourier transform which requires the signal to be stable and defined for all real numbers.

A specific case of the Laplace transform is the Fourier transform.

Fourier analysis is a term used in mathematics to describe the study of both operations.

The Fourier transform exists when the Fourier integral given below converges.

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

The Fourier transform is an integral transformation of $f(t)$ from the time domain to the frequency domain.

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega$$

The function $f(t)$ and its transform form the Fourier transform pairs:

$$f(t) \Leftrightarrow F(\omega)$$

Fourier Transform

- $F(\omega)$ is a complex function;
- Its magnitude is called the amplitude spectrum
- Its phase is called the phase spectrum
- Thus $F(\omega)$ is the spectrum
- The Fourier transform exists when the Fourier integral converges and the sufficient condition required is given below:

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty$$

Fourier Transform pairs

$f(t)$	$F(\omega)$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$u(t + \tau) - u(t - \tau)$	$2\frac{\sin\omega\tau}{\omega}$
$ t $	$\frac{-2}{\omega^2}$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$
$\delta(t - t_0)$	$e^{-j\omega t_0}$

Fourier Transform pairs

$f(t)$	$F(\omega)$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$
$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$
$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$\cos \omega_0 t$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$
$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$

Fourier Transform vs Laplace Transform

Feature 	Fourier Transform	Laplace Transform
Domain	Converts time-domain to frequency-domain ($j\omega$)	Converts time-domain to s-domain ($s = \sigma + j\omega$)
Base Functions	Uses undamped complex exponentials ($e^{j\omega t}$)	Uses decaying/growing complex exponentials (e^{-st})
Function Applicability	Limited to stable, absolutely integrable signals	Can handle a wider range of signals, including unstable ones
Analysis	Analyzes frequency components, spectrum, and energy density	Analyzes system stability, transient behavior, and solves differential equations
Relationship	Is a special case of the Laplace transform when $\sigma = 0$ and the region of convergence includes the imaginary axis	The more general transformation

Fourier Transform applied to electrical circuits

- The Fourier transform is applied to circuits with non-periodic sinusoidal excitations in the same way as phasor techniques.

$$V(\omega) = Z(\omega)I(\omega)$$

- Where $V(\omega)$ – FT of Voltage, $I(\omega)$ – FT of current and $Z(\omega)$ – Impedance.

R	$\Rightarrow$	R
L	$\Rightarrow$	$j\omega L$
C	$\Rightarrow$	$\frac{1}{j\omega C}$

Fourier Transform applied to electrical circuits

- Transform the circuit elements into frequency domain and take the Fourier transform of the excitations.
- Use any circuit solving technique to find the response in frequency domain.
- Take the inverse Fourier transform to get the response in time domain.

Next class: Module 5